DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)

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INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)

1.ABSTRACT

Sanket (संकेत — "signal") predicts where India's weather forecast will be wrong, and says why. It answers the question a duty forecaster has at 6 am — how likely is the forecast I am holding to be badly wrong today? For 666 districts, 8 surface variables and lead days 1–10, it turns the fresh NOAA GEFS ensemble into a bust probability — error above the 90th percentile of that variable's own history — and attributes every prediction with SHAP, refreshed every six hours. SIH 2026 · Problem Statement 26079 · Ministry of Earth Sciences.

2.PROBLEM STATEMENT

- Every centre issues forecasts — but nothing says when today's will be badly wrong.
- A bust is a tail error — above the 90th percentile of a variable's own error history. It is the day a warning is missed.
- Correction (NMRF/NCMRWF) removes consistent errors; a bust comes from that day's pattern, so it survives correction.
- Ensemble spread alone is a noisy signal — 5 of 31 GEFS members are published, and spread is not probability.
- Farming, disaster response and power planning across 666 districts run on forecasts that carry no confidence label.

3.PROPOSED SOLUTION

HOW A FORECAST BECOMES A VERDICT

INGEST	BUILD	PREDICT	EXPLAIN	ACT
GEFSv12 reforecast 2000–2019 + live feed; ERA5 truth; byte-range GRIB2 via .idx	666 GADM districts; area-weighted means; spread, jumps, tendencies, lagged ens. + history	XGBoost error models; bust classifier on out-of-fold errors; promotion gate	SHAP attribution; per-variable thresholds; CNN challenger scored on the same rows	map, rail, alerts, replay; held-out 2017 check; refreshed every 6 h on a 512 MB box

5.KEY FEATURES

SIGNAL, NOT SURPRISE

666
districts scored,
Day 1 to Day 10

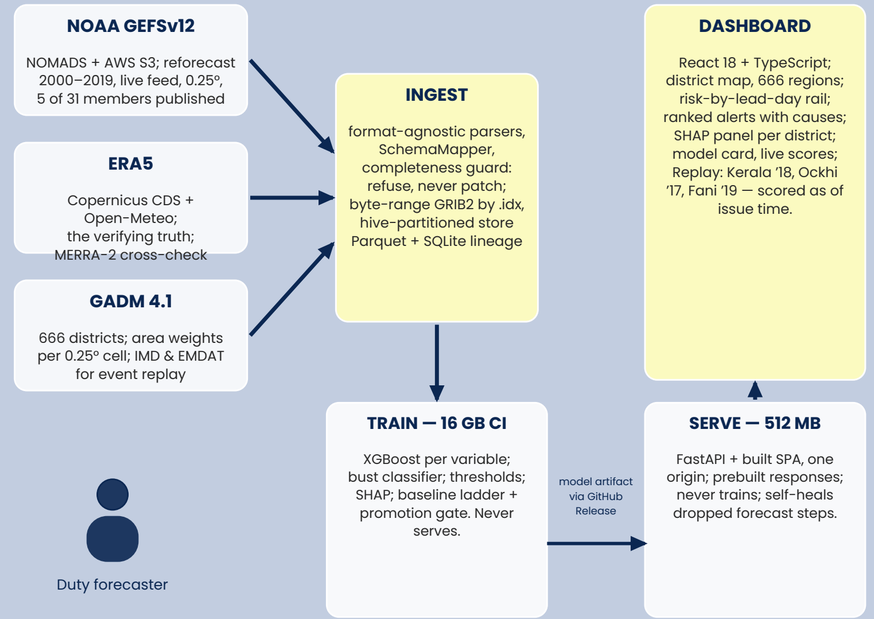
8
surface variables,
each with its own bust line

90th
percentile of error
= the bust threshold

6 h
refresh from the live
NOAA GEFS feed

Try Sanket
sanket-a0dd.onrender.com

4.ARCHITECTURE DIAGRAM



6.PROGRAM ALIGNMENT

Program outcomes (POs) — filled = mapped

P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011
●	●	●	●	●	●	●		●		●

Program specific outcomes (PSOs)

PSO 1	PSO 2	PSO 3
●	●	●

Program education objectives (PEOs)

PEO 1	PEO 2	PEO 3	PEO 4
●	●	●	●

UN Sustainable Development Goals

- 13 CLIMATE ACTION
- 9 Industry, Innovation & Infrastructure
- 11 SUSTAINABLE CITIES AND COMMUNITIES
- 10 Reduced Inequalities

TRL Level

6
Technology demonstrated in a relevant environment

7.HOW SANKET IS DIFFERENT

vs. correction (NMRF/NCMRWF): correction removes errors a model makes consistently. A bust is one day's pattern — it survives correction. Sanket predicts that residual risk, with a probability and reasons, instead of a corrected number.

vs. ensemble spread: spread is an input, not a verdict. Sanket trains on 17 years of reforecasts and scores spread, run-to-run jumps, tendencies, season and district history together, against a published ladder of baselines.

vs. another dashboard: Sanket says where the forecast is likely to break — per district, variable and lead day — and attributes every alert with SHAP. Kerala 2018, Ockhi and Fani can be replayed as of issue time. Held-out 2017; real data only, no placeholders.

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